Scuola di Ingegneria Industriale e dell'Informazione

POLITECNICO
MILANO 1863

Metrics for Measuring the Quality of Unstructured Data

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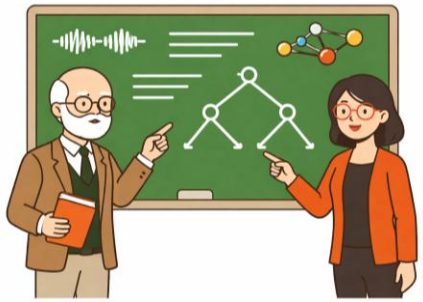
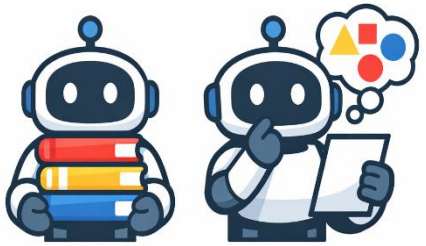
Objective

It is possible to conceptualize Data Quality measurement as a task to be learnt from data.
This work explores the definition and assessment of a quality model for text.

Theory: We consider general approaches for quality assessment in unstructured data.
Then we review the applicable dimensions for text quality.

Practice: We describe the use of classifiers as quality measurement tools.
Then we implement *model-based metrics* for the SEAHORSE dataset.





General approaches for unstructured data quality

Dimensions: Multiple perspectives [3]

- Alternative ways to define *ideal data*.
- Data pipelines involve both machine/algorithm and human consumers.

Dimension	Focus	Ideal
Interpretability	Data consumer	Data satisfies expectations • **Human consumer**: expectations determined by personal knowledge, experiences, and objectives. • **Machine consumer**: expectations defined by rules, statistical models, properties of previously observed data (training data), types, taxonomies, formats and conventions.
Relevancy	Task	Data optimizes performance
Accuracy	Real world	Data is representative • **Direct comparison:** Conditioned on the availability of a gold-standard reference dataset containing the ground-truth values as annotated by human experts who evaluated the unstructured data objects. • **Estimation:** Rules to verify conformance to expected patterns. (Similar to the consistency dimension with semantic rules [81])

Methods: Quality assessment can use analysis methods [76]

Domain-specific: Rules, parametric methods, stochastic models.
Data-driven: unsupervised and supervised methods.

Approach	Idea	Examples
Educated Guess	Form hypotheses of the data structure to generate predictions	• Hidden Markov models • Association rules
Methods in the Wild	Making no assumptions of data properties, inquire directly for its structure and quality	• Clustering • Dimensionality reduction
Neural Networks	Learn arbitrarily complex functions that involve the encoding of representations at different levels of abstraction, extracting relevant features from the data	• Self-Organized Map (SOM) • Deep Neural Networks (DNN)

Interventions in organizations: The Heterogeneous Data Quality Methodology (HDQM) [75]

Conceptual modeling separates:
- *intensional semantics* (definitions and meaning).
- *extensional semantics* (values).

Data Quality in texts

Patterns in Natural Language [39]

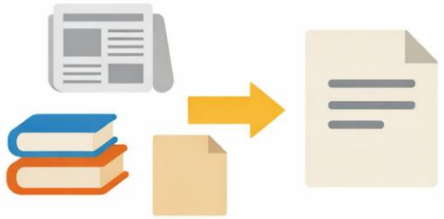
Five linguistic elements:

1. *Lexicon*: words.
2. *Grammar*: rules.
3. *Semantics*: meaning.
4. *Rhetoric*: literary effect.
5. *Pragmatics*: context.

Patterns are largely language-specific.

Dimensions[49]

- *Inherent / language quality*: text in itself.
- *Referential / content quality*: text relation to reference documents.



Quality dimensions for texts

Inherent / Language Quality dimensions [39]

- Readability (length statistical correlations)
- Comprehensibility (fit for target audience)

- Lexical Accuracy (vocabulary)
- Syntactic Accuracy (grammar rules)

- Cohesion (connection of text units)
- Coherence (creation of a mental model)

- Linguistic accessibility (words)
- Cultural accessibility (skills)

Referential / Content Quality dimensions [31, 48]

- Relevance (important and no redundant)
- Factuality (entailment)
- Adequacy (lossless, undistorted, unexpanded)

We focus on three quality dimensions defined on single texts:

- *Comprehensible*
- *Grammatical*
- *Repetition-free*

We do not consider referential quality:

- Highly task specific (e.g. translation, summarization)
- Dimensions associated to specialized research areas of NLP.

Human subjectivity for type-agnostic data quality

Collected judgments

- Uses interviews and questionnaires to measure the contextual value of data [18, 19, 20, 14].
- Relies on psychometric methods for quantification [14, 12, 16].

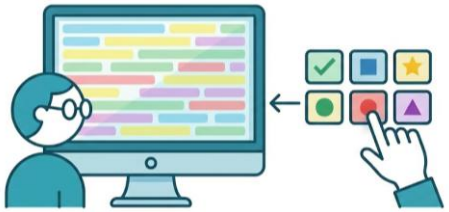
Notable implementation: *AIMQ*
Uses an instrument called the *Information Quality Assessment survey (IQA)*, based on a zero-to-ten *Likert-type* scale.

Recorded perceptions

- Some properties are difficult to measure directly, have a lack of formal scales, or involve multiple valid perspectives [27, 28, 29, 30].
- Annotation projects employ experts to assign attributes to data objects.

Expertise is multifaceted [35, 30]:

- Translation of sensory information.
- Domain-specific knowledge.
- Lived experiences and familiarity to different contexts.



Model-based data quality metrics

How to define a model-based quality metric

Create a classifier to discriminate a satisfactory quality class [3, 4].

1. Identify attributes that indicate *fitness for use*.
2. Select a dataset representative of human judgment patterns.
3. Train and evaluate a model to predict these patterns (*meta-metric* [1]).
4. Use membership predictions as quality estimations [3].

ROC-AUC: Threshold independent Ranking capacity [144, 145]
Probability that a satisfactory instance receives a higher score than a defective instance (equivalent to the Mann-Whitney U statistic).

Pearson Correlation Coefficient
Consistency with human judgment.

Requirements of data quality metrics [5]

I. Existence of minimum and maximum values. ✅
II. Interpretability of interval. ⚠️
III. Scientific criteria guide configuration and measurement. ⚠️
IV. Sound aggregation at different levels of granularity. ⚠️
V. Economic efficiency. ✅

⬇

- In *calibrated* models the interval has a probabilistic interpretation. This allows general sound aggregation.
- *Min* and *Max* operators are safe aggregations.

- Some models exhibit calibration in their train distribution.
- It is possible to apply specialized calibration methods.
- Metrics are assumed local until further validation [2, 91].

- Documentation and frameworks describe best practices.

Model-based approaches for text quality

Evaluating models: Benchmarks

- Consist of datasets, evaluation metrics and aggregation rules to compute comparable performance scores [54].
- Can evaluate specific tasks or multiple to inquire high level model capabilities (e.g. GLUE for Natural Language Understanding) [55, 54].
- Describe covered features and use cases, and are representative of real-world workloads [57].

Model based approaches for Text evaluation [48]

1. Unsupervised Matching: semantic equivalence with a reference.
2. Supervised Regression: predict human judgments → Easily aligned with fitness for use.
3. Supervised Ranking: scoring functions.
4. Text Generation: output probabilities of generative models.

Class membership predictions can serve as quality metrics [3, 4].

Families of text classification methods [53]

I. Bags-of-Words (e.g. Support Vector Machines).
II. Tokens (e.g. Transformer DNNs).
III. Hierarchies and graphs that model relationships of information (e.g. Graph Neural Networks).

Transformer Architectures [53]

- Encoder-only (masked) → SOTA
- Encoder-decoder (text-to-text)
- Decoder-only (causal)

Model selection

Selected model: Multilingual cased DistilBERT [164]

- Encoder-only architecture (SOTA).
- Pretrained on Wikipedia in 104 Languages.
- Reduced parameter count facilitates training.
- Lightweight size supports deployment:
  - Fast Inference.
  - Low computational/hardware requirements.
  - Can operate in real time on edge and mobile devices.

Can provide an *initial interface* for measuring data quality.
To increase performance, it is possible to perform additional pretraining or switch to a more complex model.

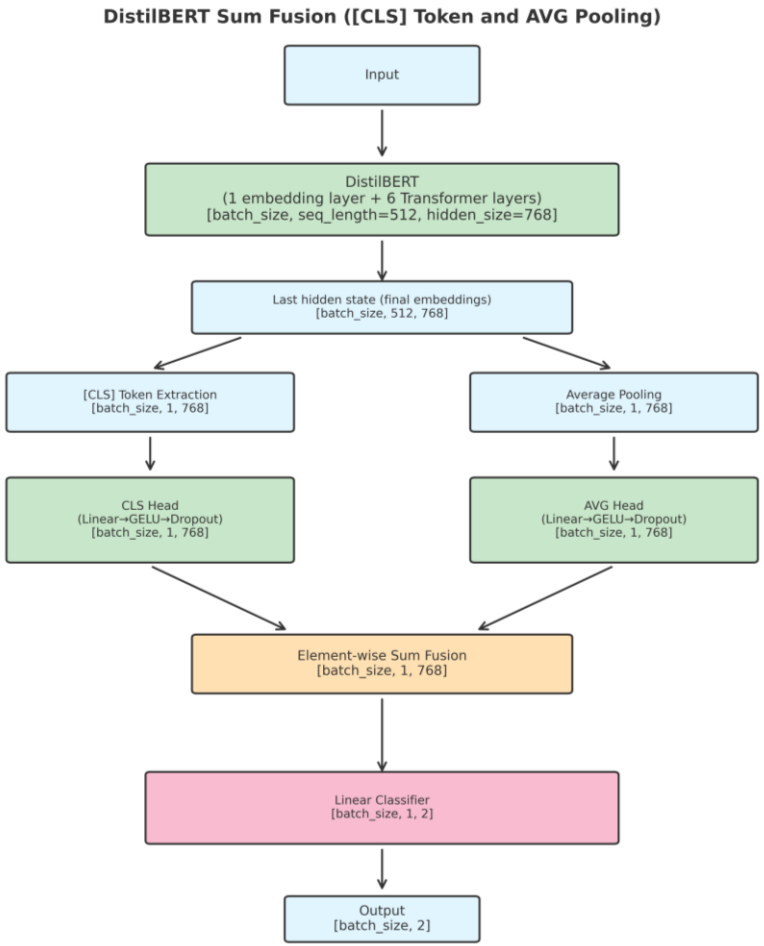
Variants

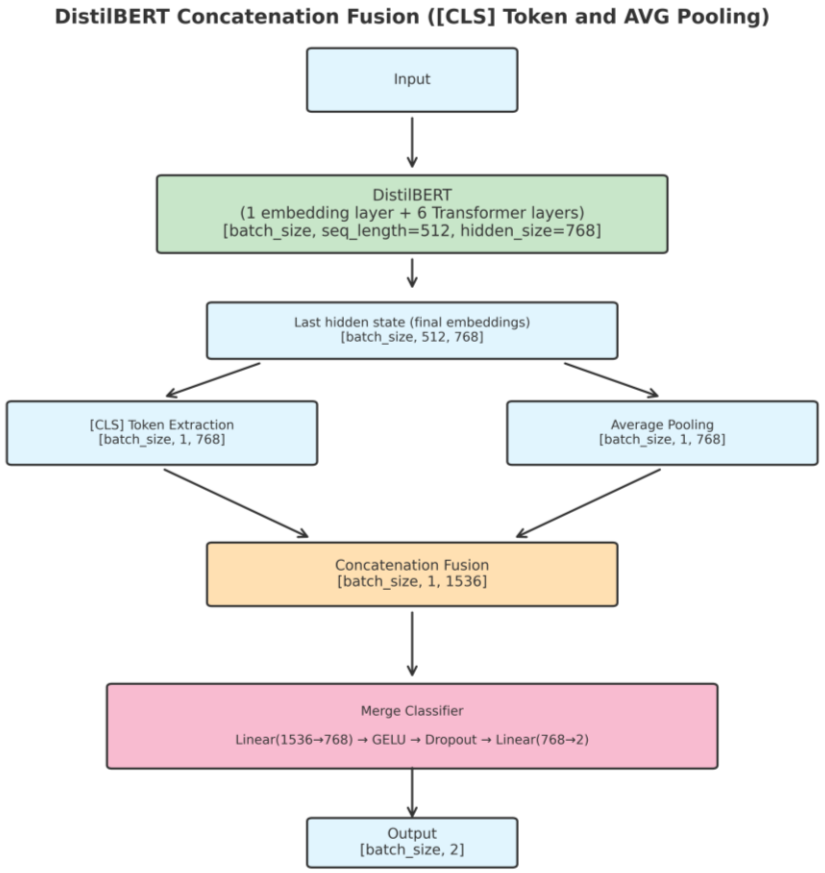
Approach	Description
Fine Tuning (library default)	The transformers library class DistilBertForSequenceClassification provides a simple straightforward model to fine-tune DistilBERT for downstream tasks. It consists of DistilBERT connected to one hidden layer and an output layer using the standard [CLS] token as representation. We consider tuning half of the transformer layers (3/6) as a baseline.
Feature extraction	Training of a classifier head that takes as input the [CLS] and average pooling sequence representations of the last 3 transformer layers. Advantages are low training cost and containing different higher and lower level features [61].
Concatenation and Sum fusion schemes	Fine Tuning of the last 2 or 3 transformer layers using classification heads that combine the [CLS] and Average Pooling representations of the input sequence as described in [187]
Low Rank Adaption (LoRA)	Efficient adaption of last 3 layers

Visualization of model Variants

Fusion schemes [187]

- Combine sequence representations using Sum and Concatenation operations.
- By default, DistilBERT uses a [CLS] Token representation for classification.
- Including a pooling vector can increase performance.

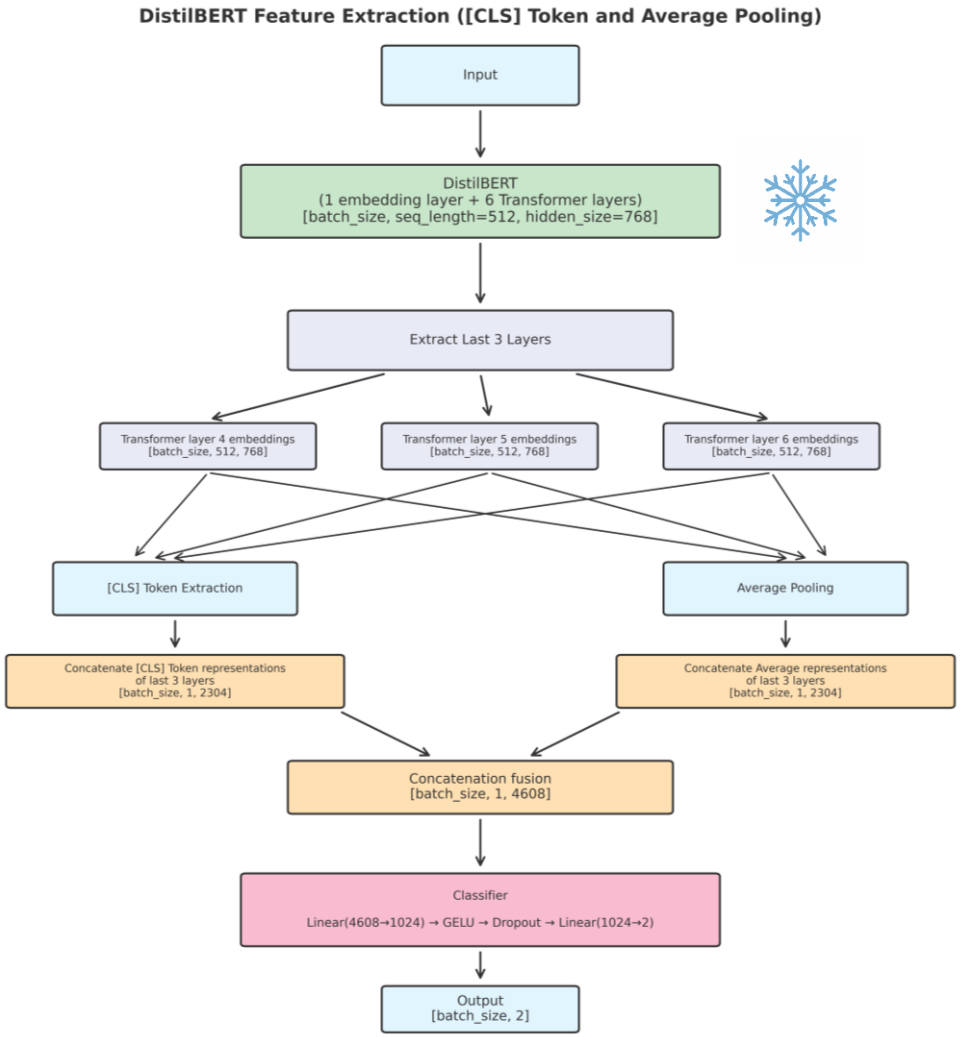




Visualization of model Variants

Feature extraction [61]

- No fine-tuning: Transformer layers are frozen and only the classifier head is trainable.
- Access the last 3 original layers. Uses features of at differet levels.



Methodology: Procedure

Procedure

1. **Data preprocessing:** Adapt the dataset to the selected model type.

2. **Initial Exploration:** For each dimension and model variant, train a classifier. Consider validation set for focusing subsequent efforts.
3. **Hyperparameter tuning:** Apply a model-specific evidence-supported heuristic. Following [132], we set the learning rate to minimize validation loss, and then consider regularization.
4. **Statistical significance test:** Study model superiority. For ROC-AUC, we use DeLong tests with Holm correction to preserve confidence.

5. **Internal Evaluation:** Evaluate models using the SEAHORSE test-splits. Compare results with those reported by the dataset creators.
6. **External evaluation:** Evaluate models using different datasets to study whether the metric is locally constrained or has wider generalization potential.

Tools: Models were implemented in Python

Libraries: Pandas, HuggingFace Datasets, HuggingFace Transformers, peft development, statsmodels, Mlstatkit.

We installed Nvidia CUDA to run experiments on the GPU device. Environment was set to be deterministic.

Class imbalance
In quality-aware organizations, defective instances constitute exceptions and not the norm.

The three considered dimensions present class imbalance (1:7).

We use weighted Cross-Entropy Loss.

Selected datasets: Training and internal validation

The SEAHORSE dataset [135]

- **Size**: 96.645 instances.
- **Languages**: English, German, Russian, Spanish, Turkish, and Vietnamese.
- **Topic**: summaries of News and Wikihow guide articles.
- **Task**: summarization.
- **Rows**: A summary text, annotations for six text quality dimensions, and an identifier to retrieve the referenced document.

Selected datasets: External validation

Repetition-free dimension

	MRPC: Microsoft Research Paraphrase Corpus [206]	**BREAD**: Boilerplate and Redundancy Evaluation on Assorted Documents [211, 204].	**PAWS-X**: Cross-lingual Paraphrase Adversaries from Word Scrambling [212, 207].
Multilingual?	No (English only)	Yes (360 languages)	Yes (7 languages)
Topic	News	Web data	Web data
Notes		Designed to test domain-driven and language-agnostic methods.	Has difficult examples characterized by repeated words but not repeated information.

Selected datasets: External validation

Comprehensible dimension

	mFACE: Multilingual Summarization with Factual Consistency Evaluation (mFACE) [195]
Multilingual?	Yes (45 languages)
Topic	News

Grammatical dimension

	MELA: Multilingual Evaluation of Linguistic Acceptability [213]
Multilingual?	Yes (10 languages)
Topic	examples for the problem of syntactic acceptability.
Notes	Distribution is not operational/natural. Linguistically curated, taking data from domain specific corpuses. Had synthetic methods to introduce error diversity.

Results: SEAHORSE Test-Set - internal validation and comparison with reference values

Dimension		*Comprehensible*		*Repetition free*		*Grammatical*	
Model	Size (Parameters)	ROC AUC	ρ Pearson correlation	ROC AUC	ρ Pearson correlation	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (3 fine-tuned layers)	135.3M	0.873211	0.470622	0.966784	0.790262	0.773207	0.313210
DistilBERT Fine tuning (2 fine-tuned layers, Concat fusion)	135.9M	0.871819	0.467516	-	-	-	-
DistilBERT Fine tuning (2 fine-tuned layers, Sum fusion)	135.9M	-	-	0.965419	0.792378	0.774605	0.341288
DistilBERT Feature extraction	139.5M	0.863720	0.465166	0.949850	0.699082	0.746517	0.297221
$mt5_L$	1.2B	0.88	0.44	0.97	0.74	0.81	0.37
$mt5_{XXL}$	13B	0.90	0.52	0.98	0.86	0.84	0.45

- **Reference results:** *mT5* encoder-decoder models trained by the dataset creators [135].
- Our model with more parameters has a ratio of 0.12 in parameter count to the lightest *mT5* model (≈ x8) .
- Best scoring models, compared to mT5 preserve 99%, 99%, and 92% of ROC-AUC . Feature extraction preserves above 88%.
- ROC-AUC values above 0.74 show good ranking performance.

Results: External validation and problems for out-of-distribution data

Comprehensible

mFACE 5 common languages with SEAHORSE results (Comprehensible)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (3 layers)	0.735731	0.215162
DistilBERT Fine tuning (2 layers, Concat fusion)	0.730868	0.205156
DistilBERT Feature extraction	0.726941	0.200867
$mt5_L$	0.77	0.14
$mt5_{XXL}$	0.73	0.09

mFACE all 45 languages results (Comprehensible)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (3 layers)	0.661992	0.138677
DistilBERT Fine tuning (2 layers, Concat fusion)	0.658755	0.133781
DistilBERT Feature extraction	0.656241	0.140323
$mt5_L$	0.68	0.13
$mt5_{XXL}$	0.70	0.15

Repetition-free

MRPC results (Repetition)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Feature extraction	0.702786	0.293978
DistilBERT Fine tuning (2 layers, Sum fusion)	0.677008	0.265046
DistilBERT Fine tuning (3 layers)	0.656483	0.250313

BREAD results (Repetition)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Feature extraction	0.936411	0.749494
DistilBERT Fine tuning (2 layers, Sum fusion)	0.879358	0.641187
DistilBERT Fine tuning (3 layers)	0.813980	0.522627

PAWS-X 3 common languages with SEAHORSE results (Repetition)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (2 layers, Sum fusion)	0.533732	0.022461
DistilBERT Fine tuning (3 layers)	0.531459	0.024834
DistilBERT Feature extraction	0.526131	0.018420

PAWS-X 7 languages results (Repetition)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (2 layers, Sum fusion)	0.526573	0.018646
DistilBERT Fine tuning (3 layers)	0.525606	0.023242
DistilBERT Feature extraction	0.523394	0.012872

Grammatical

MELA 4 common languages with SEAHORSE results (Grammatical)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (2 layers, Sum fusion)	0.516648	0.013285
DistilBERT Fine tuning (3 layers)	0.516182	0.022934
DistilBERT Feature extraction	0.506810	0.002019

MELA all 10 languages results (Grammatical)

Model	ROC AUC	ρ Pearson correlation
DistilBERT Fine tuning (2 layers, Sum fusion)	0.549429	0.045536
DistilBERT Feature extraction	0.547713	0.050555
DistilBERT Fine tuning (3 layers)	0.542532	0.052952

Retrain with MELA training set

TN = 1322 FP = 703 FN = 2370 TP = 3026

Accuracy	: 0.5859 | 4348 / 7421
F1 Score	: 0.6632
Precision	: 0.8115 | 3026 / 3729
Sensitivity (Recall, TPR)	: 0.5608 | 3026 / 5396
Specificity (TNR)	: 0.6528 | 1322 / 2025
Youden's J statistic	: 0.2136
ROC AUC	: 0.6468
Pearson correlation	: r = 0.2299 p = 1.4291×10^{-89}
MCC	: 0.1903

Conclusions and future work

Conclusions

1. High quality data can be used to train high performance models, which can serve as quality metrics (virtuous circle)

2. Lightweight models can provide a cost-effective implementation.

3. Data quality is context dependent. Data-driven tools are fit to process i.i.d data but may fail on new distributions (o.o.d data).

Potential directions for future work:

- Automatic-metrics in dynamic contexts may require monitoring, retraining and recalibration.

- Guidelines to create training datasets. For example, through small scale annotation projects or automatic annotations (silver datasets).

- Calibration approaches and methodologies for sound aggregation.

- Referential quality, involving several data objects, or using a trusted knowledge base.

Thank you for your attention

Questions are welcome.